

- per client summary and full transaction list sorted by time

1) main() starts

main()
 no local variable yet

Heap -> objects created with new empty

- JVM calls main() string args)

2) local variables initialized.
 stack - LIFO

main()
 filePath
 totalBuyAmountByClient
 totalSellAmountByClient
 transactionList

active

-> string #1

-> HashMap #1

-> HashMap #2

-> ArrayList #1

filePath, both maps, transactionList are reference variable

Heap -> objects created with new

string #1
 string
 "2 / transaction-V1.csv"

HashMap #1
 empty HashMap

HashMap #2
 HashMap empty

ArrayList #1
 empty ArrayList

3) file opened, header skipped.
 stack

main()
 filePath
 totalBuyAmountByClient
 totalSellAmountByClient
 transactionList
 bufferedReader

active

- string #1

- HashMap #1

- HashMap #2

- ArrayList #1

- Buffered Reader #1

Heap

FileReader #1
 FileReader
 open os file handle

BufferedReader #1
 BufferedReader wraps -> FileReader

4) Reading and splitting a line
 stack - LIFO

main()
 filePath
 totalBuyAmountByClient
 totalSellAmountByClient
 transactionList
 bufferedReader
 line
 values

active

- string #1

- HashMap #1

- HashMap #2

- ArrayList #1

- Buffered Reader #1

- string #2

- string[] #1

BufferedReader #1
 BufferedReader wraps -> FileReader

String #2
 string
 "T1001,C100,BUY,500,2024-01-05 9:15:00"

String[] #1
 string
 "T1001" "2024-01-05"
 "C100" "9:15:00"
 "BUY"
 "5000"

5) parsing the fields.
stack

Heap

main() active
...earlier Vars unchanged)
line → string #2
values
transactionid → "T1001"
clientid → "C100"
transactionType → "BUY"
amount 5000.0
timestamp → "2024-01-05 9:15:100"

string[] #1
string[]
elements aliased by the Vars
above - no new strings created

c) constructing the transaction - frame pushed.
stack - LIFO

Heap

transaction() active
this → transaction #1
transactionid → "T1001"
clientid → "C100"
transactionType → "BUY"
amount → 5000.0
timestamp → "2024-01-05 9:15:00"

Transaction #1
transaction
being filled by the constructor

main() active
transactionid → "T1001"
clientid → "C100"
amount → 5000.0
transaction (not yet assigned)

7) constructor returns - frame popped.
stack

Heap

main() active
...earlier Vars unchanged)
transactionid → "T1001"
amount → 5000.0
transaction → Transaction #1

Transaction #1.
transaction.
transactionId: "T1001"
clientId: "C100"
transactionType: "BUY"
amount: 5000.0
timestamp: "2024-01-05 9:15:00"

8) storing it, and updating the map stack.

Heap

main() active
 totalBuyAmountByClient → Hashmap #1
~~total~~
 transaction list → ArrayList #2
 transaction → Transaction #1

Transaction #1
 transaction
 "T1001", "C100", "Buy", 5000.0

ArrayList #1. new
 ArrayList
 [→ Transaction #1]

Double #1 new
 Double (auto boxed)
 5000.0

HashMap #1 new
 HashMap
 "C100" → Double #1

9) Loop continues for every row

Stack.

Heap

main() active
 totalBuyAmountByClient → Hashmap #1
 totalSellAmountByClient → Hashmap #2
 transaction list → ArrayList #1
 line → String #4 (3rd row)
 values → String [] #3 (3rd row)
 transaction → Transaction #3

String #2
 no longer referenced GC eligible

String [] #1
 no longer referenced GC eligible.

Transaction #1
 transaction
 "T1001", "C100", "Buy", 5000.0
g.l.s

Transaction #2 new
 transaction
 "T1002", "C200", "sell", 2000.0, 4000.0

Transaction #3 new
 transaction
 "T1003", "C100", "sell", 1000.0,
 next at 2000

ArrayList #1
 ArrayList
 [→ TX #1, → TX #2, TX #3]

HashMap #1
 HashMap (buy totals)
 "C100" → 5000.0

HashMap #2 new
 HashMap (sell totals)
 "C200" → 2000.0
 "C100" → 4000.0

to) sorting by timestamp
Stack

Heap

main() - active
transactionList
ArrayList#1

Comparator#1 new
Comparator
from transaction :: set timestamp

ArrayList#1 new
ArrayList
before: [-> TX#1 (g: 15) -> TX#2
TX#3 (next dat) (g: 15)
after:
[-> TX#2 (g: 10), TX#1 (g: 5)
-> TX#3 (next dat)]

1) Cleanup - close and return
Stack

Heap

empty
- main() has
returned

HashMap#1 GC eligible
Unreachable.

HashMap#2 - GC eligible

ArrayList - GC eligible

Transaction#1, #2, #3 - GC eligible

BufferedReader#1 - GC eligible